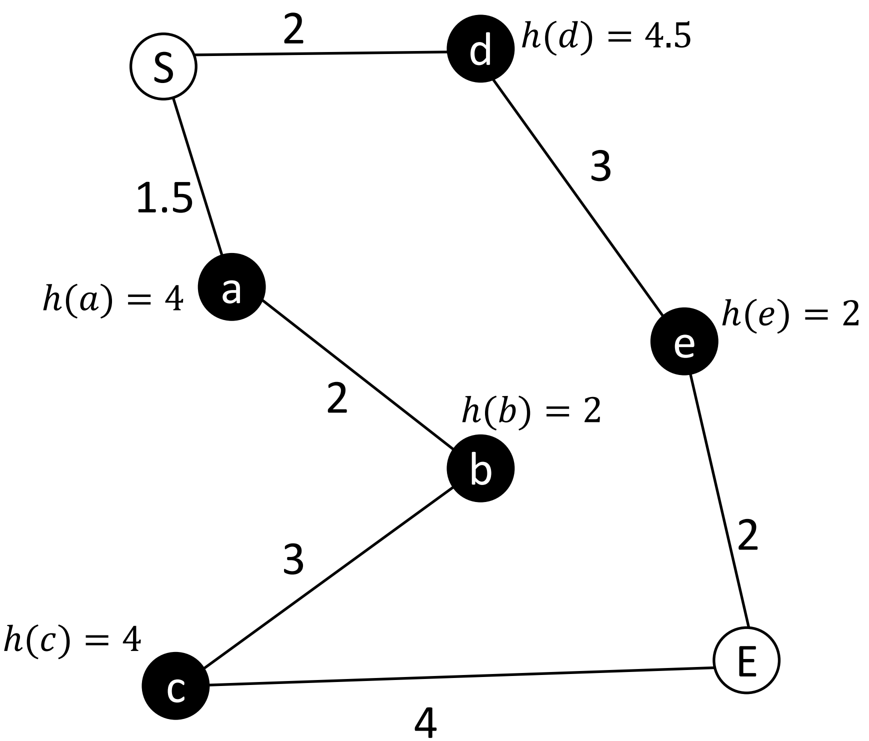
Tutorial questions

1. A* search
2. Coordinated turn and Dubins path
3. MATLAB example state estimation

Tutorial question 1a – A*

Apply A* to find the optimal path from start node S to end node E.

The cost following a path between two nodes $g(n)$ and the heuristic function $h(n)$ at each node are shown on the graph.



① $f(a) = g(a) + h(a) = 1.5 + 4 = 5.5$

$f(d) = 2 + 4.5 = 6.5$

② $f(b) = (2 + 1.5) + 2 = 5.5$

$f(d) = 6.5$

③ $f(c) = (1.5 + 2 + 3) + 4 = 10.5$

$f(d) = 6.5$

④ $f(c) = 10.5$

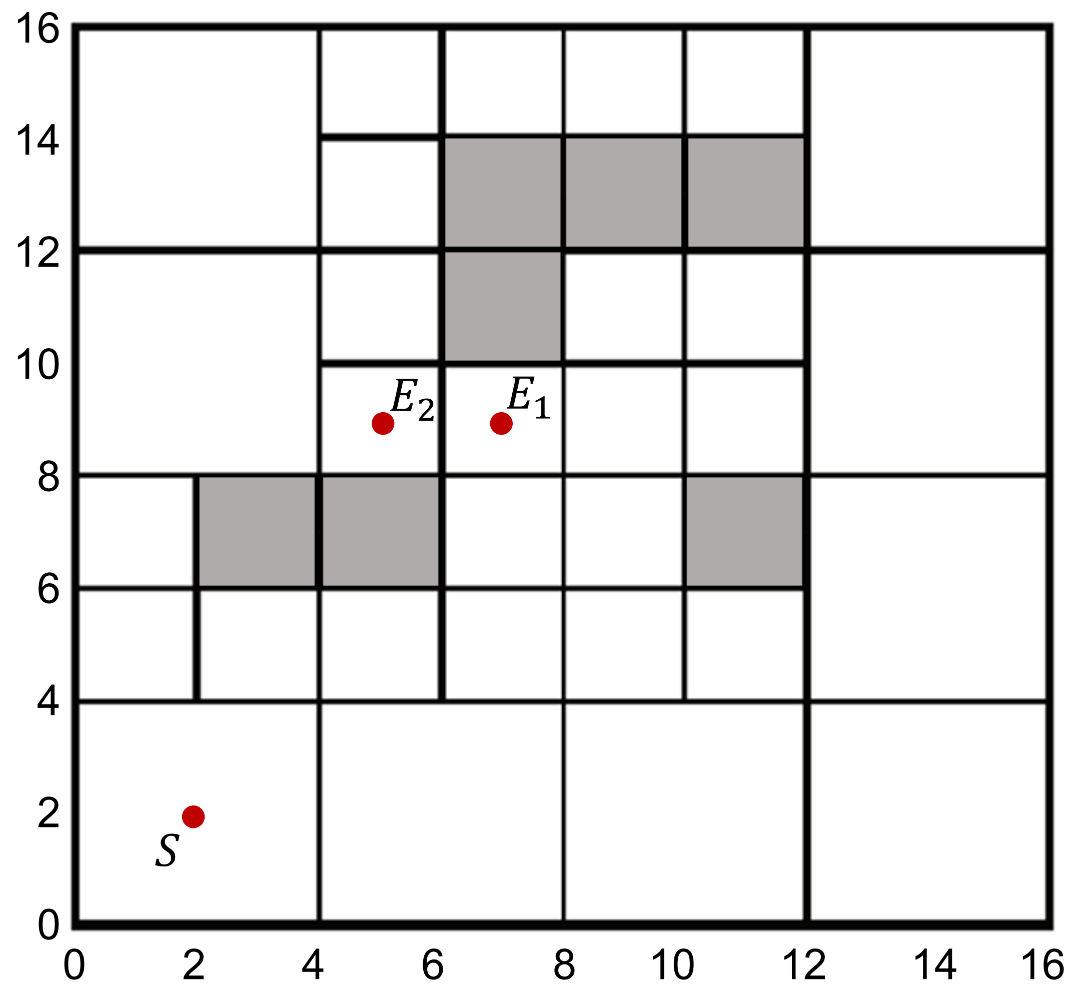
$f(e) = (2+3) + 2 = 7$

$\therefore$ select S - d - e - E

Tutorial question 1b – A*

Apply A* to find the optimal path from start node S to end node E_1 (and E_2).

- The cost following a path between two nodes is the vertical (or horizontal) distance for an adjacent move (same sized cells), 1.1 for adjacent moves between cells of different size, and 1.4 times the distance for a diagonal move.
- The heuristic function $h(n)$ at each node n is the straight-line distance between n and the end node E.



for E_1,

① $f(a) = 3 \cdot 1.1 + \sqrt{6^2 + 4^2} = 10.5$

$f(b) = 3.3 + 4\sqrt{2} = 8.96$

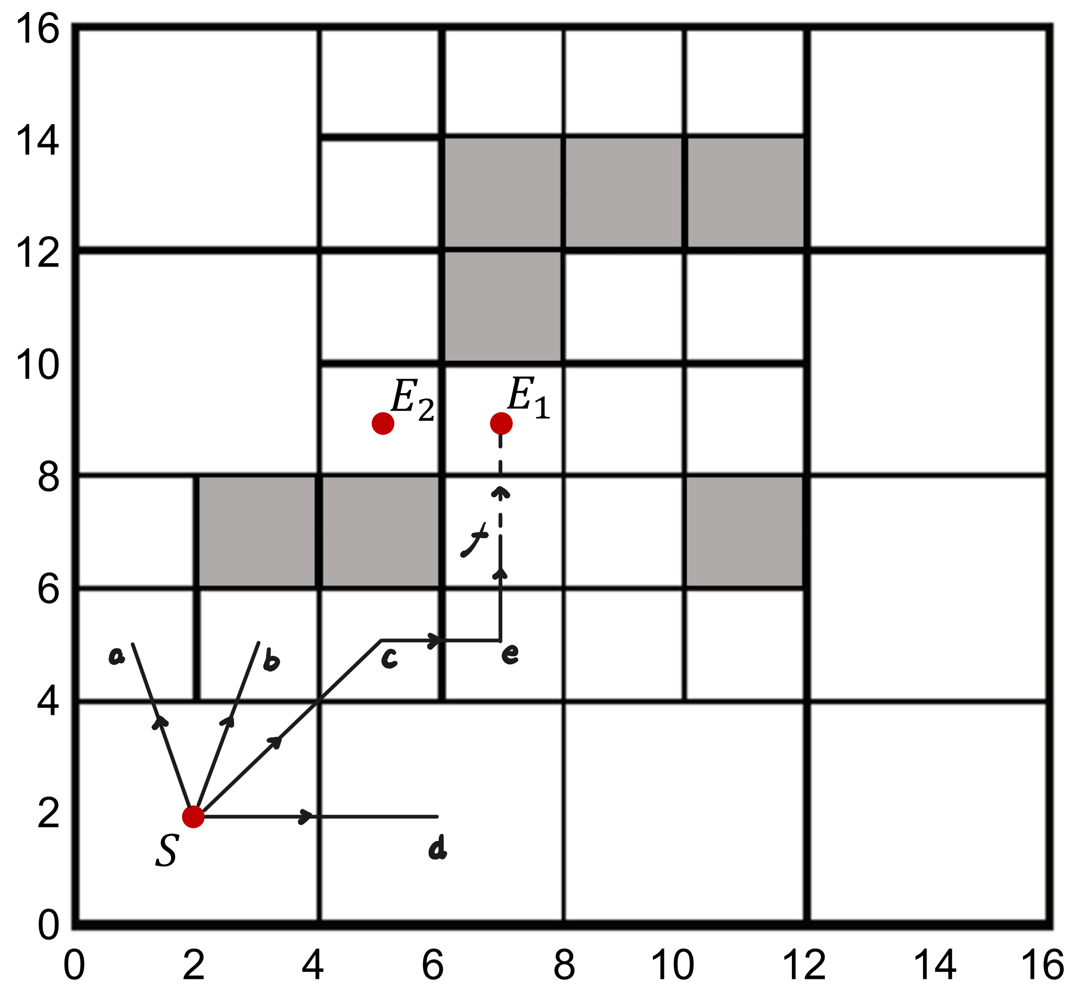
$f(c) = 4.2 + 2\sqrt{5} = 8.67$

$f(d) = 4 + 5\sqrt{2} = 11.1$

② $f(e) = 6.2 + 4 = 10.2$

③ $f(f) = 8.2 + 2 = 10.2$

$\therefore$ select $S - c - e - f - E_1$



for E_2.

① $f(a) = 3\cdot 1.1 + \sqrt{4^2+4^2} = 8.96$

$f(b) = 3.3 + 2\sqrt{5} = 7.77$

$f(c) = 4.2 + 4 = 8.2$

$f(d) = 4 + 5\sqrt{2} = 11.1$

② $f(c_b) = 5.3 + 4 = 9.3$

$f(a_b) = 5.3 + 4\sqrt{2} = 11.0$

③ $f(e) = 6.2 + 2\sqrt{5} = 10.7$

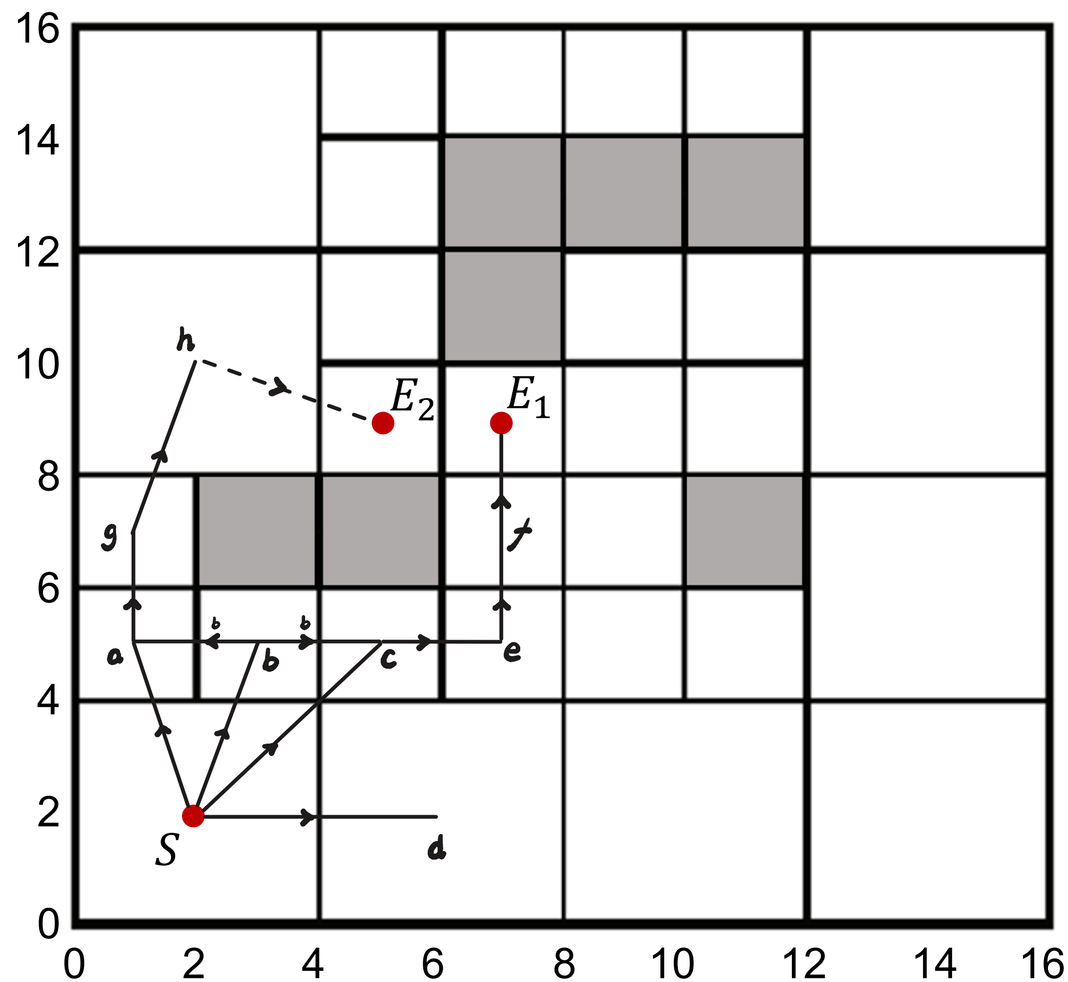
④ $f(g) = 5.3 + 2\sqrt{5} = 9.77$

⑤ $f(h) = 8.6 + \sqrt{10} = 11.8$

⑥ $f(f) = 8.2 + 2\sqrt{2} = 11.0$

⑦ $f(E_1) = 10.2 + 2 = 12.2$

∴ select $S - a - g - h - E_2$

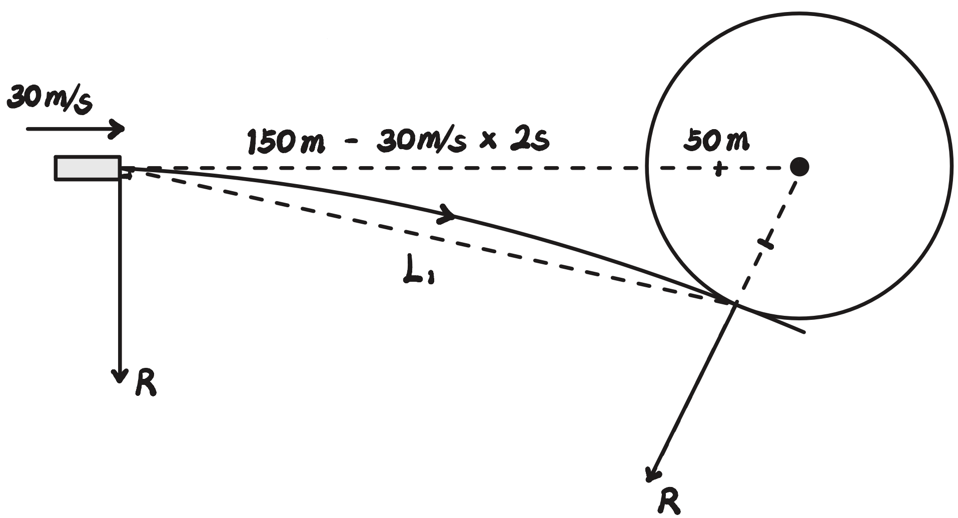


Tutorial question 2

A fixed-wing UAV flying steady level at 30 m/s towards a cylinder-shaped obstacle of radius 50 m senses the obstacle at a distance of 150 m from the obstacle wall. The UAV has a reaction time of 2 s before it can bank and get into a coordinated turn.

1. What value of the bank angle should the UAV's control system command to ensure that the UAV can just avoid colliding with the obstacle while performing a coordinated turn?
2. Suppose that the tightest turn at 30 m/s corresponds to a turn radius of 50 m. The vehicle plans a Dubins path which just avoids (grazes) the obstacle, with only the initial heading specified as above (Dubins path consisting of right turn & straight line). Ignoring the time required to recover the UAV to wings-level flight, derive (but do not solve) closed-form equations whose solution yields the grazing point.

1.

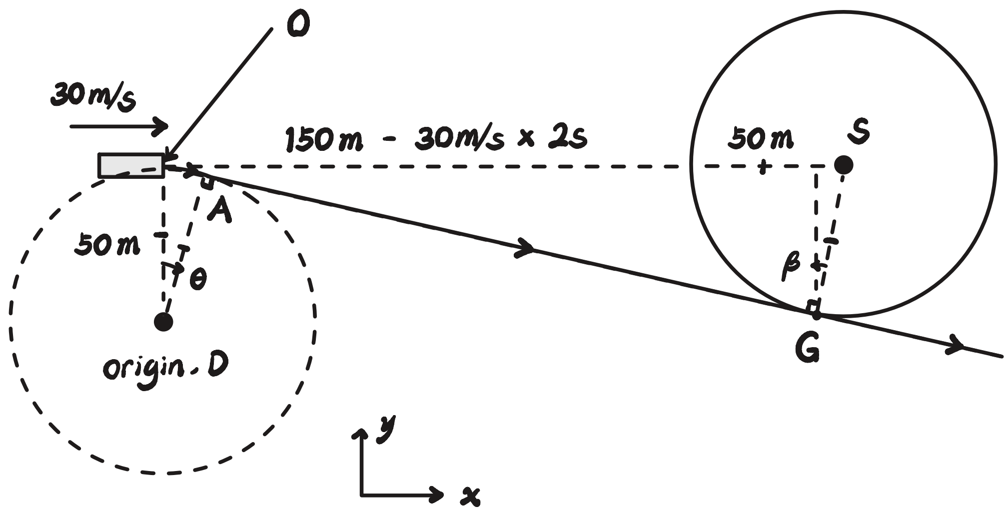


$\therefore\ (R+50)^2 = R^2 + 140^2$

$\therefore\ R^2 + 100R + 2500 = R^2 + 19600$

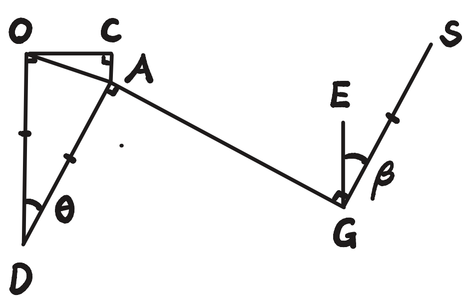
$\therefore\ R = 171\ (m)$

2.



$$x_A = 50\sin\theta \quad \text{and} \quad x_G = 140 - 50\sin\beta$$

$$y_A = 50\cos\theta \qquad\qquad y_G = 50 - 50\cos\beta$$



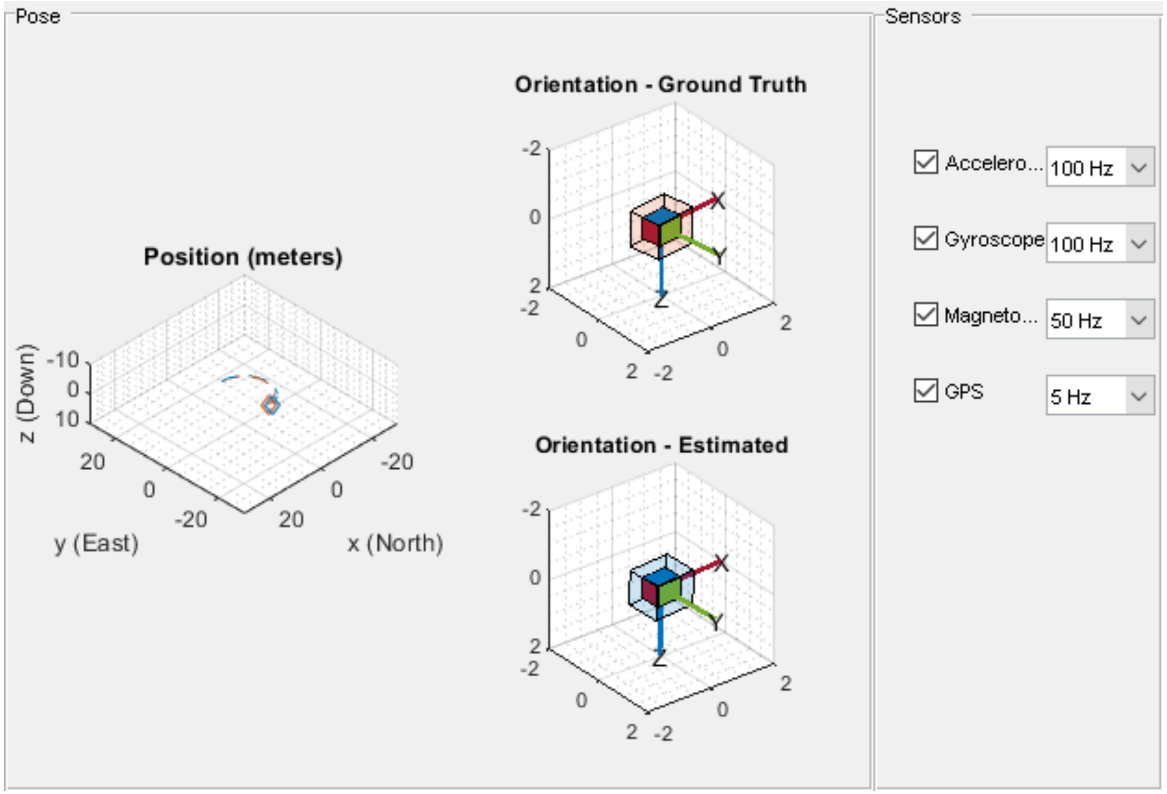
$\because AD \parallel GS,\ DO \parallel GE \quad \therefore \theta = \beta \quad (1)$

$\because DA \perp AG \quad \therefore \overrightarrow{DA} \cdot \overrightarrow{AG} = 0$

$$\therefore \begin{bmatrix} x_A \\ y_A \end{bmatrix} \cdot \begin{bmatrix} x_G - x_A \\ y_G - y_A \end{bmatrix} = x_A(x_G - x_A) + y_A(y_G - y_A) = 0 \quad (2)$$

Tutorial question 3

Explore the MATLAB state estimation tutorial, where **accelerometer, gyroscope, magnetometer and GPS sensors are fused to estimate** the orientation and position of a vehicle moving along a circular path.



MATLAB script: https://www.mathworks.com/help/fusion/ug/pose-estimation-from-asynchronous-sensors.html

Tutorial video: https://www.youtube.com/watch?v=hN8dL55rP5I